Yiming Zhang
Curriculum Vitae

Institute for Rock Magnetism
University of Minnesota, Twin Cities
Minneapolis, MN 55455
yiming-z@umn.edu
duseryiming@gmail.com
+1 (323) 810-2226
`duserzym.github.io`

ACADEMIC APPOINTMENT

Postdoctoral Associate, Institute for Rock Magnetism, University of Minnesota, Twin Cities — Sep 2024-

Postdoctoral Scholar, Earth and Planetary Science, University of California, Berkeley — May-Sep 2024

EDUCATION

Ph.D., Earth and Planetary Science, University of California, Berkeley — 2024

B.A, cum laude., Geology, Occidental College — 2019

PUBLICATIONS
(* indicates mentored student)

Zhang, Y., Hodgin, E.B., Schmitz, M.D., Schwartz, D., Mohr, M.T., Crowley, J.L, Swanson-Hysell, N.L., (2026), Protracted high-grade metamorphism and slow cooling in the Adirondacks recalibrate Neoproterozoic global paleogeography. Tectonics. `https://doi.org/10.1029/2025TC009333`

Kahn, L.X.*, **Zhang, Y**., Finnegan, Seth., Hodgin, E.B., Swanson-Hysell, N.L., (2025), The stratigraphic record of the arrival of the Sacramento and San Joaquin Rivers to the California coast. Journal of Sedimentary Research. `https://doi.org/10.2110/jsr.2024.153`

Fuentes, A.J., Courtney-Davies, Liam., Flowers, R.M., **Zhang, Y**., and Swanson-Hysell, N.L., (2025), Evolution of iron formation to ore during Ediacaran to early Paleozoic tectonic stability. Earth and Planetary Science Letters. `https://doi.org/10.1016/j.epsl.2025.119621`

Swanson-Hysell, N.L., **Zhang, Y**., Macdonald, F.A., Koran, I., Tasistro-Hart, A., and Jay, A., (2025), Oman was on the northern margin of a wide late Tonian Mozambique Ocean. Geology. `https://doi.org/10.1130/G53450.1`

Wenk, HR., Kattemalavadi, A., **Zhang, Y**., Kennedy, E.R., Borkiewicz, O., (2025), Exploring microstructures and anisotropies of serpentinites. Contributions to Mineralogy and Petrology. `https://doi.org/10.1007/s00410-025-02209-5`

Zhang, Y., Anderson, N.S.*, Mohr, M.T., Nelson, L.L., Macdonald, F.A., Schmitz, M.D., Thurston, O.G., Guenthner, W.R., Karlstrom, K.E., and Swanson-Hysell, N.L., (2024), Paleomagnetism of the southwest Laurentia large igneous province and Cardenas Basalt: pulsed magmatism during rapid late Mesoproterozoic plate motion. JGR Solid Earth. `https://doi.org/10.1029/2024JB029036`

Hodgin., E.B., Swanson-Hysell, N.L., Kylander-Clark, A.R.C., Turner, A.C., Stolper, D.A., Ibarra, D.E., Schmitz, M.D., **Zhang, Y**., Fairchild, L.M., Fuentes, A.J., (2024), One billion years of stability in the North American Midcontinent following two-stage Grenvillian structural inversion. Tectonics. `https://doi.org/10.1029/2024TC008415`

Zhang, Y., Hodgin, E.B., Alemu, T., Pierce, J.*, Fuentes, A., and Swanson-Hysell, N.L., (2024), Tracking Rodinia into the Neoproterozoic: new paleomagnetic constraints from the Jacobsville Formation. Tectonics. `https://doi.org/10.1029/2023TC007866`

Mohr, M.T., Schmitz, M.D., Swanson-Hysell, N.L., Karlstrom, K.E., Macdonald, F.A., Holland, M.E., **Zhang, Y**., Anderson, N.*, (2024), High-Precision U-Pb geochronology links magmatism in the SW Laurentia Large Igneous Province and Midcontinent Rift. Geology. `https://doi.org/10.1130/G51786.1`

Sapienza, F., Gallo, L.S., **Zhang, Y**., Vaes, Bram., Domeier, Mathew., Swanson-Hysell, N.L., (2023), Quantitative Analysis of Paleomagnetic Sampling Strategies. JGR Solid Earth. `https://doi.org/10.1029/2023jb027211`

Gallo, L. C., Domeier, M., Sapienza, F., Swanson-Hysell, N. L., Vaes, B., **Zhang, Y**., Arnould, M., Eyster, A., Guȓer, D., Király, Á., Robert, B., Rolf., T., Shephard, G., and van der Boon, A., (2023), Embracing uncertainty to resolve polar wander: a case study of Cenozoic North America. Geophysical Research Letters. `https://doi.org/10.1029/2023GL103436`

Slotznick, S.P., Swanson-Hysell, N.L., **Zhang, Y**., Clayton, K.E., Wellman, C.H., Tosca, N.J., and Strother, P.K., (2023), Reconstructing the paleoenvironment of an oxygenated Mesoproterozoic shoreline and its record of life. GSA Bulletin. `https://doi.org/10.1130/B36634.1`

Pierce, J.*, **Zhang, Y**., Hodgin, E.B., and Swanson-Hysell, N.L., (2022), Quantifying inclination shallowing and representing flattening uncertainty in sedimentary paleomagnetic poles. Geochemistry, Geophysics, Geosystems. `https://doi.org/10.1029/2022GC010682`

Rose, I., **Zhang, Y**., and Swanson-Hysell, N.L., (2022), Bayesian paleomagnetic Euler pole inversion for paleogeographic reconstruction and analysis. JGR: Solid Earth. `https://doi.org/10.1029/2021jb023890`

Zhang, Y., Swanson-Hysell, N.L., Avery, M.S., Fu, R.R., (2022), High geomagnetic field intensity recorded by anorthosite xenoliths requires a strongly powered late Mesoproterozoic geodynamo. PNAS. `https://doi.org/10.1073/pnas.2202875119`

Hodgin, E.B., Swanson-Hysell, N.L., DeGraff, J.M., Kylander-Clark, A.R.C., Schmitz, M.D., Turner, A.C., **Zhang, Y**., Stolper, D.A., (2022), Final inversion of the Midcontinent Rift during the Rigolet Phase of the Grenvillian Orogeny. Geology. `https://doi.org/10.1130/G49439.1`

Cromwell, G., **Zhang, Y**., (2021), New paleointensity data from Aniakchak volcano, Alaska, USA. Geochemistry, Geophysics, Geosystems. `https://doi.org/10.1029/2021gc010032`

Zhang, Y., Swanson-Hysell, N.L., Schmitz, M.D., Miller Jr., J.D., and Avery, M.S., (2021), Synchronous emplacement of the anorthosite xenolith-bearing Beaver River diabase and one of the largest lava flows on Earth. Geochemistry, Geophysics, Geosystems. `https://doi.org/10.1029/2021GC009909`

Swanson-Hysell, N.L., Avery, M.S., **Zhang, Y**., Hodgin, E.B., Sherwood, R.J., Apen, F.E., et al., (2021). The paleogeography of Laurentia in its early years: new constraints from the Paleoproterozoic East Central Minnesota batholith. Tectonics. `https://doi.org/10.1029/2021TC0067511`

Swanson-Hysell, N.L., Hoaglund, S.A., Crowley, J.L., Schmitz, M.D., **Zhang, Y**., and Miller Jr., J.D., (2020), Rapid emplacement of massive Duluth Complex intrusions within the Midcontinent Rift. Geology.

https://doi.org/10.1130/g47873.1

Zhang, Y., Pairing paleointensity results with coercivity spectra: providing support for selection criteria. *IRM Quarterly. Volume 30. Number 1.*

TALKS

Space Physics Seminar, University of Minnesota Twin Cities (invited) April 14, 2026
The Geodynamo Through Time: core paradoxes, paleomagnetic insights, and open questions

Geology Seminar, Occidental College (invited) November 4, 2025
Deep time, moving continents, and a snowball Earth

Division Seminar, Caltech (invited) November 3, 2025
Assembling supercontinent Rodinia and tracking its development in the late Proterozoic

2025 GSA Annual Meeting October 19-22, 2025
Recalibrating the history of metamorphism and exhumation in the Adirondack Highlands leads to a major revision of Neoproterozoic global paleogeography

2025 GSA Annual Meeting (invited) October 19-22, 2025
Paleogeographic Evolution of Oman and Egyptian Terranes of the Arabian-Nubian Shield through the Neoproterozoic

Earth & Environmental Sciences Department seminar, UMN Duluth April 8, 2025
Anorthosites in northern Minnesota: 100 years of study of their 1-billion-year-old history

Scripps Institute for Oceanography October 7, 2024
Reconstructing late Proterozoic magmatism, geomagnetic field behavior, and paleogeography by pairing paleomagnetism with state-of-the-art geochronology

Hard Rock Lunch seminar, University of Minnesota Twin Cities October 2, 2024
Reconstructing Neoproterozoic paleogeography with metamorphic rocks

UC Berkeley EPS exit seminar April 25, 2024
Reconstructing late Proterozoic magmatism, geomagnetic field behavior, and paleogeography using tiny magnets in rocks

2023 AGU Fall Meeting Dec 5 2023
Constraining the position of Arabian-Nubian arc terranes in the lead-up to Snowball Earth glaciation: new constraints from Tonian dikes of Oman

Institute of Geology and Geophysics seminar, Chinese Academy of Science (*invited)* *July 5 2023*
New approaches to APWP synthesis and incorporating uncertainties in sedimentary records

2023 IRM conference (*invited)* *June 5-June 8 2023*
New approaches to apparent polar wander path development constrain the late Mesoproterozoic assembly of Rodinia

2023 MagIC workshop (*invited)* *Feb 28-Mar 2 2023*
New perspectives on Laurentia's Grenville Loop: tracking Rodinia across the Mesoproterozoic to Neoproterozoic boundary

2022 AGU Fall Meeting Dec 12-16 2022

Reconstructing the position of the supercontinent Rodinia in the early Neoproterozoic: new constraints from Laurentia's interior and the Grenville margin

Beijing Paleomagnetism and Geochronology Laboratory (*invited, online*) *Sep 28 2022*
High geomagnetic field intensity recorded by anorthosite xenoliths requires a strongly powered late Mesoproterozoic geodynamo

Young CEED 21 Frontiers in quantitative paleogeography (*invited, online*) *Nov 14-20 2021*
Bayesian_PEP_inversion: a Bayesian framework for integrating paleomagnetic and geochronologic data into apparent polar wander inversion

Grand Canyon Supergroup Field Forum (*invited*) *April 9-19 2021*
The rich paleomagnetic record of the Mesoproterozoic Midcontinent Rift and the Southwestern Laurentia Large Igneous Province

Cloud Meeting on Paleomagnetism (*online*) *1/29/2021*
Intense magmatic activity and a strong geomagnetic field—a study on the anorthosite xenoliths hosted in the Mesoproterozoic Midcontinent Rift diabase

2020 AGU Conference (*online*) *12/15/2020*
Recovering Mesoproterozoic geomagnetic field intensity using anorthosite xenoliths hosted in Midcontinent Rift diabase

North Central GSA Conference, Duluth, MN (*online*) *05/18/2019*
The rich paleomagnetic records of Proterozoic Midcontinent Rift intrusives: an updated synthesis with a new pole from the Beaver River diabase

Institute for Rock Magnetism, University of Minnesota, Minneapolis, MN (*invited*) *01/09/2019*
Paleomagnetism and rock magnetism on the Beaver River diabase and anorthosite xenoliths therein

TEACHING

Lecturer at the IRM summer school July 20226
micromagnetic modeling & magnetic microscopy

Teaching assistant for lab sessions at the IRM summer school June 2024

Graduate student instructor (GSI) for EPS 101 Field Geology and Digital Mapping Fall 2023
Advisor: Matthew Gleeson

Reader for EPS 88 PyEarth: A Python Introduction to Earth Science Spring 2023
Advisor: Nicholas Swanson-Hysell

GSI for EPS 101 Field Geology and Digital Mapping Fall 2022
Advisor: Nicholas Swanson-Hysell

Reader for EPS 115 Stratigraphy and Earth History Spring 2022
Advisor: Eben Blake Hodgin

GSI for EPS 101 Field Geology and Digital Mapping Fall 2021
Advisor: Nicholas Swanson-Hysell

GSI for EPS 50 The Planet Earth Spring 2021

Advisor: Daniel Stolper

Participant in the Graduate Remote Instruction Innovation Fellows Program — Winter 2020

GSI for EPS 101 Field Geology and Digital Mapping — Fall 2020
Advisor: Nicholas Swanson-Hysell

Completion of UC Berkeley GSI Conference training — Jan 2020

Completion of required Online Course: Professional Standards and Ethics for GSIs — Fall 2019

Completion of required Pedagogy Course EPS 375 — Fall 2019

ORIGINAL GEOLOGICAL FIELD WORK

Southern Sweden, [1 week] — 2026
Reconstructing the paleogeographic position of Baltica in the Neoproterozoic

Central Eastern Desert, Egypt [2 weeks] — 2025
Reconstructing the Neoproterozoic paleogeography of the Arabian-Nubian shield

Sliderock Mountains, Montana [1 week] — 2024
Extending site-based apparent polar wander path for North America into the Mesozoic with paleomagnetic data from the intrusive rocks of the Sliderock Mountain Volcanics

Oman [5 weeks] — 2024, 2026
Constraining the paleogeographic configuration of the Mozambique Ocean in the Neoproterozoic

Grenville orogen [3 weeks] — 2022, 2023
Pairing paleomagnetic data and thermochronology records to reevaluate the exhumation history of rocks of the Grenvillian Orogeny and recalibrate the Grenville Loop

Death Valley, California; Grand Canyon, Arizona [4 weeks] — 2021
Using paleomagnetism and geochronology to study the temporal and magmatic relationship between the ca. 1.1 Ga Southwestern Laurentia Large Igneous Province and the Midcontinent Rift

Pikes Peak, Colorado [2 weeks] — 2020
Using paleomagnetism and geochronology to study the emplacement history of Pikes Peak batholith and its temporal and magmatic associations with the Midcontinent Rift 1.1 billion years ago

Midcontinent Rift, Lake Superior Region [11 weeks] — 2019, 2020, 2021
Reconstructing the behavior of the Mesoproterozoic geomagnetic field, continental motion during supercontinent assembly and the nature of ancient environments through paleomagnetic studies of the intrusive rocks and sediments of the 1.1 billion-year-old Midcontinent Rift

Central Highland, Iceland [3 weeks] — 2019
Qualitatively and quantitatively measure the erosion rate of rofabards soil erosion in Central highlands, Iceland

AWARDS AND FUNDING

IAGA Early Career Presentation Awards — 2025

AGU GPE Post-doc grant ($755) 2024
Enhancing site-based apparent polar wander paths with Bayesian stratigraphic age models

UC Berkeley graduate student conference travel grant ($900) 2023
2023 Geochronology Gordon Research Conference

UC Berkeley Earth and Planetary Science George C. Louderback Award 2023

2023 AGeS3-Grad Geochronology Award ($8,865) 2023
Dating the Grenville Loop using U-Pb apatite thermochronology

2022 GSA Graduate Student Research Grant ($1,749) 2022
Paleomagnetism and thermochronology of the Adirondack Mountains, Grenville Province

UC Berkeley graduate student conference travel grant ($900) 2022
2022 AGU Fall Meeting oral presentation

Hearts to Humanity Eternal (H2H8) Programs ($10,000) 2022
H2H8 Association Graduate Research Grant to Advance Humanity through Science

U.S. Visiting Student Fellowships, Institute for Rock Magnetism 2019
Paleomagnetism and rock magnetism study on Mesoproterozoic Beaver Bay Complex and anorthosite xenoliths therein ($500)

ILSG Student Research Fund, Institute on Lake Superior Geology 2019
To study the emplacement history of the Beaver River diabase and the anorthosite xenoliths therein using paleomagnetism ($500)

Chevron-Xenel Gateway Fellowship, Berkeley International House ($5,000) 2019

John Parke Young Student Grant, Occidental College 2019
Using UAV Data for Monitoring Modern Rofabard Soil Erosion in Central Highlands, Iceland ($3,500)

Independent research, Occidental College 2018
Pseudo-Thellier Paleointensity Measurement on R-N Geomagnetic Polarity Reversal Recorded by Mafic Lava Flows, Anahola, Kauai ($4,000)

Independent research, Henry Luce Foundation, Nanjing University 2017
Mapping of Ambient Ozone Pollution in China and the Assessment of Its Health Impact on Socio-Economy ($2,250)

SERVICE

Contributing developer to the open source PmagPy software project (`https://github.com/PmagPy/PmagPy`).

Contributing developer to the open source RockmagPy software project (`https://github.com/PmagPy/RockmagPy-notebooks`).

Lead developer for the RapidPy superconducting rock magnetomter automatic sample changing system control software (`https://duserzym.github.io/RAPID/`).

IN THE NEWS

PNAS In This Issue:Evidence of strong Proterozoic magnetic field
https://www.pnas.org/doi/10.1073/iti2922119

EOS news: Swinging Strength of Earth's Magnetic Field Could Signal Inner Core Formation https://eos.org/articles/swinging-strength-of-earths-magnetic-field-could-signal-inner-core-formation

UC Berkeley EPS department news: EPS Research on Earth's Ancient Magnetic Field Featured in EOS:
https://eps.berkeley.edu/news/eps-research-earths-ancient-magnetic-field-featured-eos

Diamonds Are a Paleomagnetist's Best Friend in EOS:
https://eos.org/articles/diamonds-are-a-paleomagnetists-best-friend

Reviewer for the following journals

Earth and Planetary Science Letters
Geochemistry, Geophysics, Geosystems
GEOLOGY
Nature Communications Earth & Environment
Science Bulletin
Tectonics
Science Advances

Reviewer for the following panels

NASA Emerging Worlds

MEMBERSHIPS

American Geophysical Union (AGU)
Geological Society of America (GSA)